

Compared to What? A Human-Anchored Security Benchmark for LLM-Generated Infrastructure-as-Code

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Abstract—Large language models are increasingly used to author Infrastructure-as-Code (IaC), where a single insecure default is provisioned directly into a production cloud environment. Prior evaluations report absolute vulnerability counts for model-generated IaC, but without a human reference they cannot say whether models are actually *worse* than the engineers they assist. We present GenIaC-SecBench, a benchmark of 100 deployment scenarios stratified by architectural complexity, evaluated across 12 model configurations spanning four vendors and both open and closed weights, yielding 1,196 generated artifacts scanned by three independent policy engines (Checkov, Trivy, KICS) at complete coverage. Crucially, we scan a corpus of 634 human-authored IaC templates with the identical toolchain, providing the first size-matched human security baseline for this task. We find that vulnerability density is strongly inverse to artifact size (Spearman $\rho = -0.55$, $p < 10^{-77}$), so unmatched comparisons measure artifact size rather than security. Matched on declared-resource count, *every* model configuration falls in a narrow band of $3.21\times$ – $3.87\times$ the human vulnerability density, and the gap widens as tasks get simpler ($4.9\times$ at one resource, $1.4\times$ at twenty or more). We further decompose “reasoning” into three arms—standard generation, prompt-engineered chain-of-thought, and the vendor’s own extended-thinking API—and show that vendor extended thinking significantly outperforms prompted chain-of-thought (-12.0% , $p = 0.0013$), while prompted chain-of-thought alone is indistinguishable from standard generation (-1.3% , n.s.). Instrumenting token usage reveals why the effect is bounded: extended thinking consumes under 1% of the output budget on this task class. We also report two negative results: the intuitive hypothesis that more deployable models are more vulnerable is unsupported ($r = 0.158$, $p = 0.625$), and classical complete-case Friedman testing is *uncomputable* on realistic benchmark designs, motivating the Skillings–Mack statistic. All code, data, and regeneration scripts are released.

Index Terms—infrastructure as code, large language models, software security, empirical software engineering, static analysis, cloud security

I. INTRODUCTION

Infrastructure-as-Code (IaC) tools such as Terraform, AWS CloudFormation, Azure Resource Manager, and Kubernetes manifests declare cloud infrastructure as version-controlled artifacts. A misconfiguration in an IaC template is not a latent code defect: it is provisioned, and the resulting storage bucket, security group, or IAM role is exposed exactly as written. Security smells in IaC are both common and consequential [1], [2].

Large language models now generate substantial volumes of such code. A body of work has established that LLM-generated *application* code contains vulnerabilities at non-

trivial rates [3]–[5], and that developers using AI assistants may write less secure code while believing the opposite [6]. Evaluations specific to IaC are newer and thinner [7].

A recurring limitation unites these studies: they report absolute or normalized vulnerability counts *for models only*. Reporting that a model averages eight findings per declared resource invites the immediate question *compared to what?* Without a human reference measured through the same toolchain, such numbers cannot distinguish three very different worlds: models being genuinely careless, models simply emitting more infrastructure per file, or a scanner ruleset that flags any template heavily. This paper supplies the missing anchor.

Contributions.

- 1) **A size-matched human security baseline.** We scan 634 human-authored IaC templates with the same three engines and show that density must be compared within resource-count strata. Matched, every model configuration sits at $3.21\times$ – $3.87\times$ the human baseline (Section IV-D).
- 2) **A three-arm decomposition of “reasoning.”** Separating prompt-engineered chain-of-thought from vendor extended-thinking APIs shows the two are not interchangeable for security (Section IV-H).
- 3) **Direct measurement of reasoning engagement.** Extended thinking consumes $<1\%$ of output tokens on IaC, bounding the achievable effect (Section IV-H).
- 4) **A methodological correction for incomplete benchmark designs.** Complete-case Friedman retains zero usable blocks here; we adopt Skillings–Mack [8] (Section IV-E).
- 5) **Calibration of an LLM judge against three human experts,** delimiting where automated scenario review is trustworthy (Section IV-O).
- 6) **Two negative results,** reported as findings rather than omitted (Section IV-P).

II. BACKGROUND AND RELATED WORK

A. Security of LLM-generated code

Pearce et al. [3] found roughly 40% of Copilot completions in security-relevant contexts contained weaknesses. User studies show developers with AI assistance produce less secure code while reporting higher confidence [4], [6]. Large-scale

comparisons across models confirm that security varies substantially by model [5]. Benchmarks such as SecurityEval [9] and correctness suites such as EvalPlus [10] target application code, where the unit of analysis is a function. IaC differs: the unit is a declarative resource graph, correctness is schema validity, and the security question is about defaults rather than control flow.

B. IaC security

Rahman et al. [1] catalogued seven recurring security smells in IaC scripts; subsequent work extended detection across languages [11] and studied practitioner behaviour [2], [12]. Policy engines including Checkov [13], Trivy [14], and KICS [15] operationalize such rules, commonly mapped to CIS Benchmarks [16]. The closest prior evaluation of *generated* IaC is Vargas et al. [7], which evaluates text-to-Terraform generation. Our study differs in scope—four IaC formats rather than one, complexity stratification, three independent engines, multi-rater human validation—and, most importantly, in providing a human security baseline rather than model-only counts.

C. Reasoning modes

Chain-of-thought prompting improves performance on reasoning-heavy tasks [17], [18], though recent evidence suggests gains concentrate in mathematical and symbolic domains [19]. Vendors now expose reasoning as a first-class API parameter with an explicit token budget [20]–[22]. These are distinct mechanisms: one changes the prompt, the other allocates computation. To our knowledge no prior security evaluation separates them—and, as we report in Section VI, conflating them is easy to do accidentally.

D. LLM-as-a-judge

Model-based evaluation is widely used [23], [24] but exhibits known biases, including preference for self-generated content [25]. We therefore treat our judge as a secondary signal and calibrate it against three human experts.

III. METHODOLOGY

A. Scenarios

The benchmark comprises 100 natural-language deployment scenarios: 60 *simple* (single-service tasks) and 40 *complex* (multi-component architectures spanning networking, identity, data, and observability). Scenarios specify functional requirements only and never mention security controls, so that generated security posture reflects model defaults. Coverage spans AWS, Azure, GCP, and provider-agnostic Kubernetes across Terraform HCL, CloudFormation, ARM, and Kubernetes manifests.

B. Model configurations

Table I lists the 12 configurations. Beyond nine distinct models across four vendors, three configurations isolate reasoning mode on a fixed base model: *standard*, *-cot* (a chain-of-thought system-prompt suffix), and *-thinking* (the vendor

TABLE I
MODEL CONFIGURATIONS. THE THREE CLAUDE OPUS 4.6 ROWS DIFFER ONLY IN REASONING MODE, ISOLATING THAT VARIABLE.

Configuration	Vendor	Reasoning mode
claude-opus-4-6	Anthropic	none
claude-opus-4-6-cot	Anthropic	prompted CoT
claude-opus-4-6-thinking	Anthropic	extended thinking (adaptive)
claude-sonnet-4-6	Anthropic	none
gpt-5	OpenAI	none
gpt-5-thinking	OpenAI	reasoning_effort=high
gpt-4o	OpenAI	none
gemini-3.1-pro	Google	none
gemini-3.7-flash	Google	none
llama3 / mistral / phi3	open	none (local)

extended-thinking API). All calls are stateless with a fixed system prompt requesting code-only output.

C. Generation protocol

We record the protocol verbatim, since generation details materially affect security posture and are frequently underspecified in this literature [26].

Every request is a single stateless API call: no conversation history, no few-shot examples, no retrieval, and no shared memory between scenarios. The system prompt establishes the role (“senior cloud infrastructure engineer”), requests production-ready IaC, and constrains output to a single fenced code block with no surrounding commentary. The user prompt is generated from the scenario record as a target-format instruction (e.g. “Write Terraform HCL code for AWS to: *[scenario text]*”). No prompt mentions security, hardening, compliance, or any specific control, so the observed posture reflects model defaults rather than instruction-following.

Sampling temperature is fixed at 0.2 for standard and prompted-CoT arms. Extended-thinking arms use the vendor default of 1.0, which the API mandates when reasoning is enabled—a confound we cannot eliminate without disabling the feature under test, and which we note as a limitation. The chain-of-thought arm appends a single instruction to the system prompt directing step-by-step reasoning before the code block, and changes nothing else.

Output token ceilings are set well above the observed maximum: complex scenarios have a measured median of 19,562 output tokens, and truncated responses (*finish_reason* indicating the limit was reached) are *discarded* rather than written, since a truncated template remains syntactically parseable and would silently deflate both validity and finding counts. Four complex scenarios could not be completed even at the model maximum of 128k output tokens and are reported as missing.

Requests are retried on transient failures with exponential backoff. Per-request token usage, including reasoning tokens, is logged for the analysis in Section IV-H.

D. Validation and scanning

Generated artifacts are schema-validated (terraform validate, cfn-lint, ARM parsing, kubeconform), then scanned by three independent engines from three vendors—Checkov [13], Trivy [14], and KICS [15]—giving a convergent-validity argument no single ruleset provides. We report *complete* coverage: 1,196 of 1,196 artifacts scanned by all three engines, verified by a machine-readable coverage manifest regenerated on every run.

E. The human reference corpus

We draw 634 human-authored IaC templates from three public repositories of production-style and reference infrastructure. Files are filtered by structural heuristics (root keys such as `Resources:` or `apiVersion:`) to exclude non-IaC content. This corpus is scanned with the *identical* toolchain and configuration as the generated artifacts, so any ruleset bias applies equally to both sides of the comparison.

F. Metrics

Our primary metric is *vulnerability density*: total findings across all three engines divided by the number of declared resources, obtained by parsing each artifact’s syntax tree. Density normalizes for the fact that a template declaring fifty resources has more opportunity to be flagged than one declaring two. Resource counts are taken from the AST parse rather than from scanner output, which degrades silently when a file cannot be parsed (Section VI).

G. Statistical procedure

Models are compared within complexity strata using the Skillings–Mack statistic [8], the generalization of the Friedman test [27] to incomplete block designs; Section IV-E explains why the classical test is unusable here. Post-hoc pairwise comparisons use Wilcoxon signed-rank tests [28] with Holm correction [29], following Demšar [30]. Counts are additionally modelled with a negative binomial GEE [31], [32] using $\log(\text{resource_count})$ as an exposure offset, so coefficients are per-resource incidence rate ratios. Negative binomial rather than Poisson is required empirically: the observed variance-to-mean ratio is 130.0. Human–model comparisons use Mann–Whitney U [33]; structural distributions are compared with two-sample Kolmogorov–Smirnov tests [34].

IV. RESULTS

A. Corpus

Generation yielded 1,196 of a possible 1,200 artifacts (99.7%). Four complex scenarios could not be generated within the model’s maximum output window even at 128k tokens—itsself a finding about the scale of architecture these prompts elicit. Scanning produced 38,803 findings (Checkov 14,017; KICS 14,033; Trivy 10,753).

TABLE II
VULNERABILITY DENSITY BY CONFIGURATION AND STRATUM (MEAN / MEDIAN). `PHI3` DECLARES ALMOST NO PARSEABLE RESOURCES IN THE SIMPLE STRATUM, SO ITS DENSITY IS UNDEFINED THERE.

Configuration	Simple		Complex	
	mean	median	mean	median
claude-opus-4-6	12.12	4.00	4.38	0.68
claude-opus-4-6-cot	11.97	4.00	1.90	0.65
claude-opus-4-6-thinking	10.53	3.83	4.06	0.65
claude-sonnet-4-6	8.84	3.37	5.66	0.49
gemini-3.1-pro	11.24	4.40	5.55	1.65
gemini-3.7-flash	9.50	4.00	4.94	1.17
gpt-4o	12.30	6.88	6.64	1.71
gpt-5	9.45	3.25	5.49	1.25
gpt-5-thinking	8.44	3.00	5.65	1.00
llama3	11.29	4.00	4.61	1.39
mistral	10.98	5.00	8.49	5.57
phi3	—	—	4.33	4.33
Human baseline	2.66 mean / 1.00 median (n=634)			

B. Descriptive statistics

Table II reports mean and median density per configuration and stratum. Medians are markedly lower than means throughout—the distribution is heavily right-skewed and zero-inflated, which is why all inferential tests below are rank-based or explicitly negative-binomial. Fig. 1 shows the full distributions.

C. Density must be size-matched

Vulnerability density falls steeply with artifact size (Fig. 2): in the human corpus it declines from 4.51 findings per resource in single-resource files to 1.10 in files declaring twenty or more, and across generated artifacts the association is strong and highly significant (Spearman $\rho = -0.55$, $p = 1.7 \times 10^{-78}$). Because the corpora differ in scale—human files average 5.31 declared resources, simple generations ≈ 3 , complex generations ≈ 50 —any unmatched comparison measures artifact size rather than security posture. All comparisons below are therefore computed within resource-count strata.

D. LLM-generated IaC is consistently less secure than human IaC

Fig. 3 compares generated and human artifacts within matched strata. The generated corpus exhibits significantly higher density in every stratum except the largest, with the ratio decreasing monotonically in artifact size: $4.9\times$ at one declared resource, $2.4\times$ at two, $2.3\times$ at three to five, $1.8\times$ at six to ten, $2.3\times$ at eleven to twenty, and $1.4\times$ (not significant, $p = 0.058$) at twenty or more.

Aggregating per configuration over shared strata (Fig. 4) yields a strikingly narrow band: every configuration lies between $3.21\times$ and $3.87\times$ the human baseline. This consistency across four vendors, open and closed weights, and three reasoning modes suggests a property of current LLM-generated IaC rather than of any individual model.

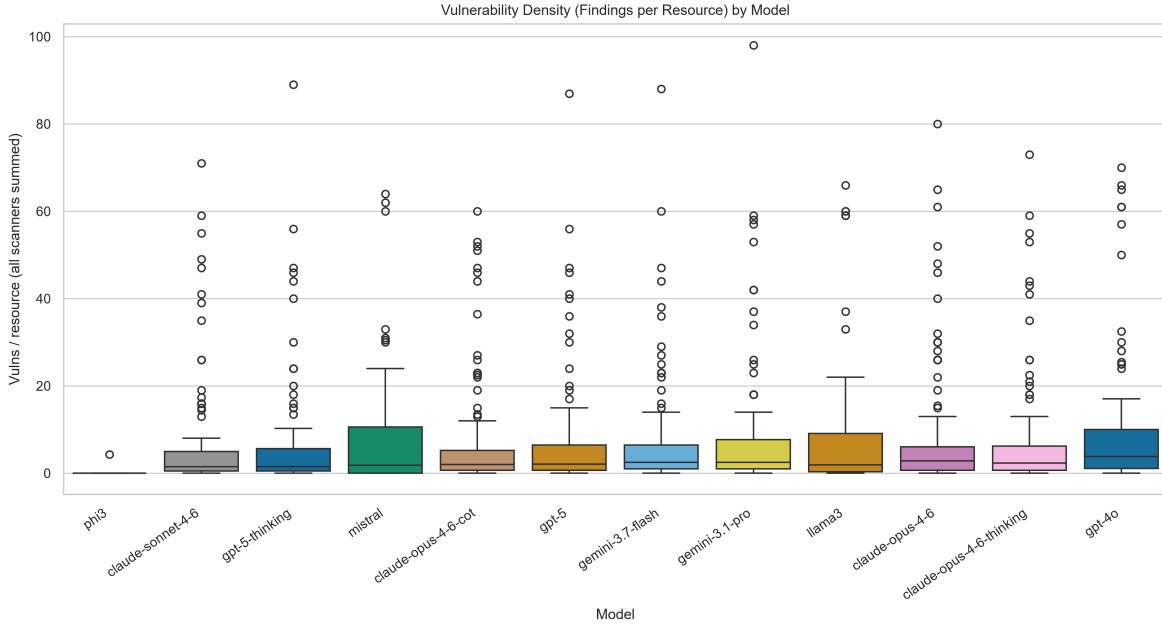


Fig. 1. Density distributions by configuration. The right skew motivates rank-based inference.

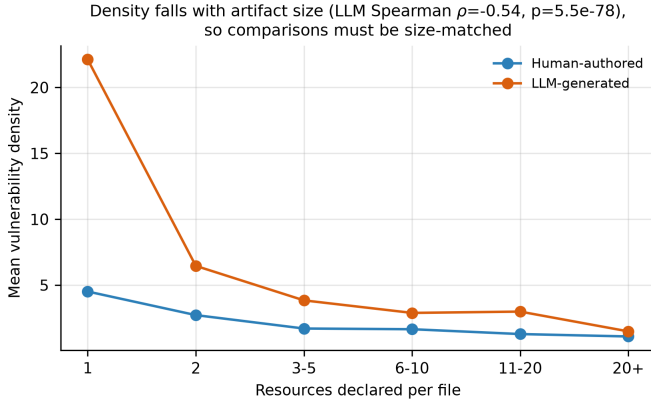


Fig. 2. Vulnerability density declines with artifact size in both corpora, requiring size-matched comparison.

Notably, the gap is *largest on the simplest tasks*. Where a human writes a minimal single-resource template, models emit substantially more flagged configuration—consistent with the structural over-generation reported in Section IV-N.

E. Models differ, but the classical test cannot show it

Skillings–Mack rejects the null of equal model performance in both strata (simple: $\chi^2 = 69.3$, $df = 10$, $p = 6.0 \times 10^{-11}$, 60 blocks; complex: $\chi^2 = 81.2$, $df = 11$, $p = 8.7 \times 10^{-13}$, 40 blocks).

The classical alternative fails outright. Friedman requires complete blocks, and with 12 configurations and realistic coverage gaps *no scenario has every configuration present*: complete-case analysis retains **zero** blocks in both strata. This is not a matter of reduced power—the standard test in this

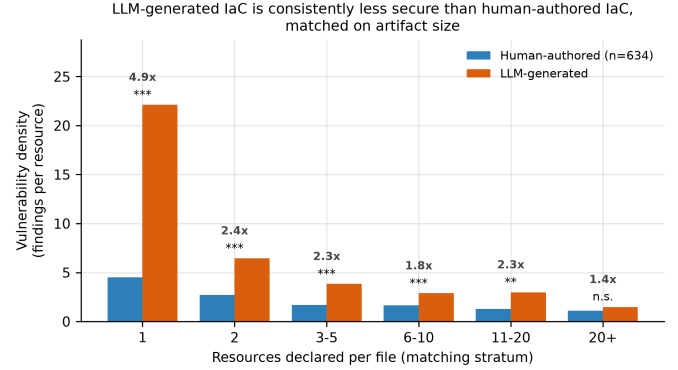


Fig. 3. Size-matched comparison against the human baseline. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (Mann–Whitney U).

literature [30] is simply uncomputable on a design of this shape. We regard this as a transferable methodological point: multi-model benchmarks routinely lose cells to refusals, quota limits, and truncation, and should default to incomplete-block statistics.

F. Post-hoc pairwise comparisons

Following Demšar [30], a significant omnibus result is followed by pairwise Wilcoxon signed-rank tests [28] with Holm correction [29]. Of 66 pairs per stratum, 10 survive correction in the simple stratum and 9 in the complex stratum (Table III). Pairwise deletion is used rather than listwise, so each comparison uses every scenario where both configurations are present; n is reported per pair.

Two patterns are visible. In the simple stratum, gpt-4o and gpt-5-thinking sit at opposite ends and separate from the

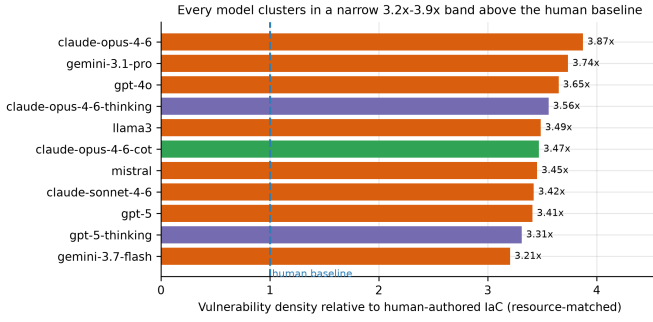


Fig. 4. Per-configuration density relative to the human baseline. Purple: vendor extended thinking. Green: prompted chain-of-thought.

TABLE III

PAIRWISE COMPARISONS SURVIVING HOLM CORRECTION ($\alpha = 0.05$). 66 PAIRS TESTED PER STRATUM.

Configuration A	Configuration B	n	p_{adj}
<i>Simple stratum (10 of 66 significant)</i>			
claude-opus-4-6	gpt-5-thinking	58	0.0036
claude-opus-4-6	claude-sonnet-4-6	60	0.0174
claude-opus-4-6-cot	gpt-5-thinking	58	0.0026
claude-opus-4-6-cot	gpt-5	59	0.0076
claude-opus-4-6-cot	claude-sonnet-4-6	60	0.0159
gemini-3.1-pro	gpt-5-thinking	58	0.0026
gemini-3.7-flash	gpt-4o	60	0.0188
gpt-4o	gpt-5-thinking	58	0.0026
gpt-4o	gpt-5	59	0.0179
claude-sonnet-4-6	gpt-4o	60	0.0448
<i>Complex stratum (9 of 66 significant)</i>			
claude-opus-4-6-thinking	gemini-3.1-pro	30	0.00065
claude-sonnet-4-6	gpt-5	30	0.0041
claude-sonnet-4-6	gemini-3.7-flash	32	0.0043
claude-opus-4-6-thinking	gemini-3.7-flash	31	0.0055
claude-sonnet-4-6	gemini-3.1-pro	31	0.0060
claude-opus-4-6	gemini-3.1-pro	32	0.0081
claude-opus-4-6-cot	gemini-3.1-pro	35	0.0112
claude-sonnet-4-6	gpt-5-thinking	29	0.0395
claude-opus-4-6	gemini-3.7-flash	33	0.0440

middle of the field. In the complex stratum, the separations are dominated by the Gemini configurations, which differ significantly from every Claude configuration—the strongest single result being `claude-opus-4-6-thinking` versus `gemini-3.1-pro` ($p_{adj} = 6.5 \times 10^{-4}$). Notably, no comparison between the three reasoning arms of the same base model survives correction, consistent with the modest within-model effects reported in Section IV-H: cross-vendor differences are larger than reasoning-mode differences.

G. Complexity interaction

The rate model includes a configuration $\times$ complexity interaction, which tests whether a configuration’s per-resource rate shifts disproportionately between strata. Table IV reports the interaction terms.

Five configurations show significant interactions, all with IRR below 1, indicating that their simple-stratum rate is *lower*

TABLE IV

CONFIGURATION $\times$ COMPLEXITY INTERACTION TERMS FROM THE NEGATIVE BINOMIAL GEE. IRR < 1 INDICATES A LOWER SIMPLE-STRATUM RATE RELATIVE TO THE CONFIGURATION’S OWN COMPLEX BASELINE.

Configuration	IRR	p
gpt-5-thinking	0.55	0.00040
llama3	0.61	0.037
gpt-5	0.63	0.00024
gpt-4o	0.64	0.275
mistral	0.64	0.637
gemini-3.7-flash	0.66	0.00092
gemini-3.1-pro	0.70	0.000067
claude-sonnet-4-6	0.59	0.164
claude-opus-4-6-thinking	0.99	0.955

TABLE V

PAIRED REASONING-MODE CONTRASTS. EVERY CONTRAST POINTS IN THE SAME DIRECTION IN BOTH STRATA; THE COMPLEX STRATUM IS UNDERPOWERED ($n \leq 33$).

Contrast	Stratum	n	Δ density	p
standard $\rightarrow$ ext. thinking	simple	60	−13.2%	0.012
	complex	31	−14.2%	0.176
prompted CoT $\rightarrow$ ext. thinking	simple	60	−12.0%	0.0013
	complex	31	−16.9%	0.209
standard $\rightarrow$ prompted CoT	simple	60	−1.3%	0.238
	complex	33	−4.8%	0.751
gpt-5 $\rightarrow$ effort=high	simple	58	−10.2%	0.153
	complex	29	+2.0%	0.701

than their own complex baseline once exposure is accounted for—most strongly `gpt-5-thinking` (IRR 0.55, $p = 4 \times 10^{-4}$) and `gpt-5` (IRR 0.63, $p = 2 \times 10^{-4}$). The Anthropic extended-thinking arm shows essentially no interaction (IRR 0.99, $p = 0.955$): its behaviour is stable across complexity.

We emphasise the direction here because the pre-remediation analysis reported the opposite for one configuration—an apparent “complexity relaxation” effect in which `gpt-4o` became 14 $\times$ more vulnerable on simple tasks. That estimate came from a model without an exposure offset and against an empty reference cell (Section VI); with both corrected, the `gpt-4o` interaction is not significant ($p = 0.275$).

H. Reasoning modes

Table V reports paired within-model contrasts on the simple stratum, where sample size is largest. Vendor extended thinking reduces density relative to standard generation (−13.2%, $p = 0.012$) and, more informatively, relative to prompted chain-of-thought (−12.0%, $p = 0.0013$). Prompted chain-of-thought alone is statistically indistinguishable from standard generation (−1.3%, $p = 0.238$).

The complex stratum reproduces the direction and magnitude of every Anthropic contrast—indeed the extended-thinking effects are slightly larger there (−14.2% and

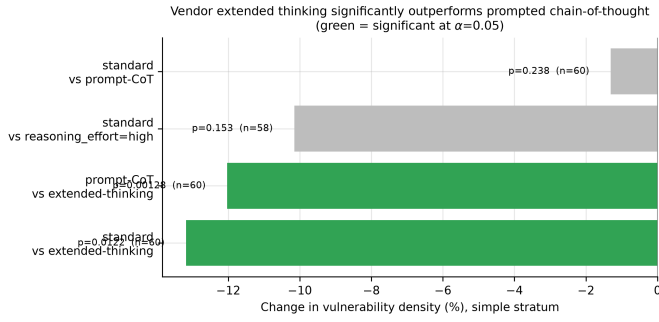


Fig. 5. Paired reasoning-mode contrasts on the simple stratum. Green bars are significant at $\alpha = 0.05$.

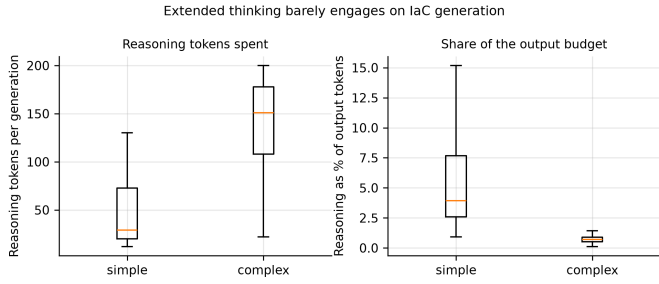


Fig. 6. Reasoning-token expenditure under extended thinking. The mechanism is barely engaged on IaC generation.

−16.9%)—but with roughly half the paired observations, none reach significance. We report these as directionally consistent but underpowered rather than as null results; distinguishing the two would require a larger complex stratum, which we identify as future work.

Fig. 5 visualizes these contrasts. The practical implication is direct: instructing a model to “think step by step” does not confer the security benefit of paying for reasoning tokens. Given that prompted chain-of-thought is often treated as a free substitute, and that its benefits appear concentrated in mathematical and symbolic domains [19], this distinction matters for practitioner guidance.

Instrumenting the API explains why the effect, though real, is bounded. Fig. 6 shows reasoning-token expenditure: a median of 29 tokens on simple scenarios and 151 on complex ones, against median completion lengths of 886 and 18,533 tokens respectively—under 1% of the output budget on complex tasks, despite a generous configured allowance. The budget is a ceiling the model may underspend, not a target. A modest measured effect should therefore *not* be read as “reasoning does not help”; on this task class the mechanism is barely exercised.

I. Rate model

The negative binomial GEE with exposure offset yields incidence rate ratios from 0.73 to 4.64 against a `claude-opus-4-6` reference. The prompted-CoT arm is significantly below the reference (IRR 0.73, 95% CI

[0.54, 0.99], $p = 0.041$); `phi3` is highest (IRR 4.64, [3.12, 6.89], $p = 2.9 \times 10^{-14}$).

The `phi3` result inverts a naive reading. It records the fewest absolute findings and appears safest by raw count, but declares almost no parseable infrastructure; *per resource* it is the worst configuration measured. This is survivorship bias made explicit, and it is why absolute counts should not be used for cross-model comparison.

J. What models get wrong, and what the engines disagree about

Mapping findings to CIS Benchmark categories [16] (Fig. 7) shows Networking as the largest named category (4,554 findings), followed by Identity and Access Management (2,849), Logging/Monitoring (2,149), and Encryption (2,140). A large residual “Other” bucket (27,111) is dominated by Checkov rules that do not map cleanly onto CIS categories and is a tooling artifact rather than a vulnerability class. The prominence of networking and IAM is consistent with the security smells catalogued for hand-written IaC [1]: permissive ingress rules and over-broad role grants remain the dominant failure modes whether the author is human or machine.

Fig. 8 shows per-engine finding counts. The three engines disagree substantially in volume—Checkov and KICS report comparable totals while Trivy reports roughly 23% fewer—which is precisely the convergent-validity argument for using three independent vendors. A single-engine study would inherit that engine’s rule coverage as if it were ground truth. We note that our own earlier, partially-covered corpus would have supported different conclusions about inter-engine agreement, which is why we report coverage explicitly.

K. Schema validity

Schema-validity pass rates (Fig. 9) cluster between 27% and 35% for all frontier configurations and collapse for the small local models (`mistral` 8%, `llama3` 6%, `phi3` 5%). The absolute rates are low because complex scenarios demand multi-service architectures that must satisfy real provider schemas; they are nonetheless directly comparable across configurations, and they are what makes the survivorship correction in Section IV-P necessary.

L. Severity distribution

Trivy and KICS emit CVSS-style severity tiers; Checkov does not without a commercial subscription, so all 14,017 Checkov findings carry UNKNOWN severity and are excluded from this analysis. Across the remaining findings the distribution is 778 CRITICAL, 3,725 HIGH, 7,919 MEDIUM, 10,036 LOW, and 2,328 INFO.

Table VI reports CRITICAL+HIGH counts by configuration. `claude-opus-4-6` leads (573), followed by its CoT (503) and extended-thinking (473) arms—an ordering that mirrors the within-model reasoning effect while remaining far above the smaller models. `phi3` records zero severity-tiered findings, again because it emits almost no parseable infrastructure rather than because it is secure.



Fig. 7. Findings by CIS category and configuration. “Other” reflects rules without a clean CIS mapping and should not be read as a vulnerability class.

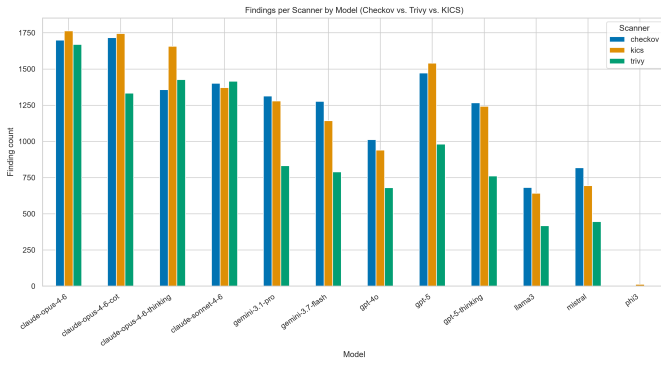


Fig. 8. Findings by engine. Volume disagreement across three independent vendors motivates multi-engine evaluation.

TABLE VI
SEVERITY-TIERED FINDINGS (TRIVY + KICS ONLY).

Configuration	Crit.	High	Med.	Low	C+H
claude-opus-4-6	124	449	971	1517	573
claude-opus-4-6-cot	92	411	908	1319	503
gpt-5	84	400	800	935	484
claude-opus-4-6-thinking	92	381	826	1432	473
gemini-3.7-flash	78	342	723	674	420
claude-sonnet-4-6	83	330	865	1400	413
gemini-3.1-pro	66	325	745	747	391
gpt-5-thinking	65	312	775	707	377
gpt-4o	46	327	556	519	373
mistral	28	280	377	367	308
llama3	20	168	372	419	188
phi3	0	0	1	0	0

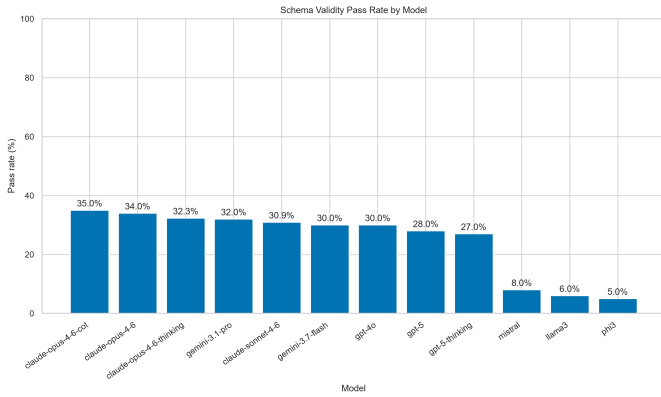


Fig. 9. Schema-validity pass rate by configuration.

These are *absolute* counts and therefore confounded by output volume; they are reported for completeness and comparability with prior work, and the per-resource analysis in Sections IV-D and IV-H remains the basis for cross-configuration claims.

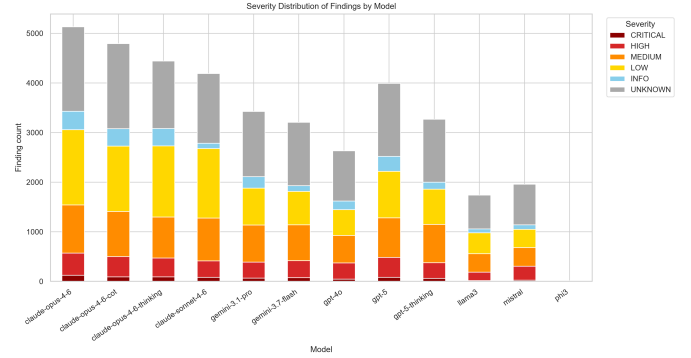


Fig. 10. Severity distribution by configuration (Trivy + KICS).

M. IaC format

Table VII reports density by target format. We report these figures with an explicit caveat: the resource-counting parser is format-dependent, and its mean counts differ sharply across formats (Terraform 23.17, ARM 2.31, Kubernetes 1.15, CloudFormation 0.90). Because resource count is the *denominator*, cross-format density values are **not** comparable, and the large apparent densities for Kubernetes and CloudFormation reflect

TABLE VII
DENSITY BY IAC FORMAT. CROSS-FORMAT VALUES ARE NOT COMPARABLE: THE RESOURCE-COUNT DENOMINATOR IS PARSER- AND FORMAT-DEPENDENT.

Format	n	mean resources	density
Terraform	836	23.17	4.60
ARM	120	2.31	3.23
CloudFormation	120	0.90	18.66 [†]
Kubernetes	120	1.15	29.11 [†]
<i>Human corpus</i>			
.tf	235	2.29	2.75
.json	188	7.06	2.65
.yaml	198	7.18	2.62

[†] inflated by an under-counted denominator.

under-counted denominators rather than dramatically worse security.

The Terraform subset—836 of 1,196 artifacts, and the format where the parser is most reliable—is the sound basis for cross-format inference, and the human corpus is likewise dominated by Terraform and CloudFormation. Within the human corpus, per-format densities are stable (2.62–2.75 for the three well-populated formats), which suggests the human baseline is not itself driven by format mix. Establishing per-format model comparisons on equal footing requires a format-normalized resource counter, which we leave to future work.

N. Structural divergence

Two-sample KS tests reject distributional equality with human-authored IaC for every configuration on every structural metric (36/36 tests, all $p < 0.05$; Table VIII). Generated templates declare 13–36 resources on average against 5.31 for humans. gpt-4o is closest to human structure across all three metrics ($D = 0.160$ – 0.189), and is the only configuration whose resource-count divergence is marginal rather than overwhelming ($p = 0.021$ against $p < 10^{-4}$ elsewhere). phi3 is most divergent ($D = 0.990$), reflecting near-total generation failure rather than a stylistic difference.

Models systematically over-generate infrastructure relative to human engineers. This matters for the central result: because density normalizes by resource count, over-generation alone does not explain the security gap in Section IV-D—models emit both more infrastructure *and* more findings per unit of infrastructure.

O. Human evaluation and judge calibration

Three cloud-security practitioners independently scored a stratified sample of scenarios on four criteria. Inter-rater agreement (Fleiss’ κ [35]) was fair on real-world plausibility (0.391) and hallucination flagging (0.266), slight on architectural coherence (0.210), and negligible on security-test relevance (0.059). Under the Landis–Koch scale [36] the last is barely above chance: experienced engineers do not agree on whether a scenario poses a meaningful security decision. This is a substantive result about the difficulty of human security evaluation, not merely a limitation of our panel.

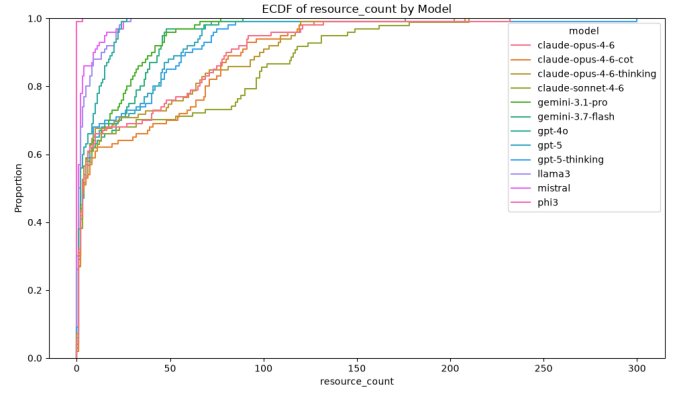


Fig. 11. Empirical CDFs of declared-resource count. Generated distributions sit to the right of the human corpus throughout.

TABLE VIII
KOLMOGOROV–SMIRNOV D AGAINST THE HUMAN CORPUS. ALL 36 TESTS REJECT AT $p < 0.05$.

Configuration	AST depth	Resource count	Diversity
gpt-4o	0.189	0.160	0.161
llama3	0.190	0.260	0.254
gemini-3.1-pro	0.219	0.246	0.231
gpt-5-thinking	0.229	0.251	0.234
claude-opus-4-6	0.259	0.292	0.268
gemini-3.7-flash	0.259	0.275	0.261
gpt-5	0.259	0.265	0.247
claude-opus-4-6-thinking	0.273	0.274	0.264
mistral	0.295	0.300	0.294
claude-opus-4-6-cot	0.309	0.335	0.308
claude-sonnet-4-6	0.322	0.301	0.306
phi3	0.990	0.990	0.984

Against the human consensus, the LLM judge is substantial on hallucination detection (94.4% exact, Cohen’s $\kappa = 0.640$ [37]) and near-perfect on plausibility (quadratic-weighted $\kappa = 0.795$, 100% within one point), but near chance on architectural coherence ($\kappa = 0.177$, 27.8% exact). The boundary is actionable: automated judges are usable for factual verification, not for architectural assessment.

P. Two negative results

First, the intuitive *validity–security* hypothesis—that models better at producing deployable code produce more vulnerable code—is unsupported. Across configurations, schema-validity pass rate and vulnerability density are uncorrelated ($r = 0.158$, $p = 0.625$; Spearman $\rho = 0.098$, $p = 0.761$). Pass rates cluster between 27% and 35% for all frontier configurations with no accompanying density relationship. Only the survivorship component holds: phi3 passes 5% and produces almost no parseable infrastructure.

Second, as reported in Section IV-E, complete-case Friedman is uncomputable on this design.

Q. Complexity stratification is measurable in the artifacts

The simple/complex split is a design choice and should be validated rather than assumed. KS tests comparing each configuration’s simple-stratum output against its own complex-stratum output reject equality for the structural metrics across configurations, confirming that the two tiers elicit materially different architectures rather than differing only in prompt wording. Separately, KS tests of each configuration against the pooled remainder find 30 of 48 tests significant at $\alpha = 0.05$, indicating that configurations occupy distinguishable structural niches. `gpt-5` and `gpt-5-thinking` are the least distinguishable from the field ($D = 0.134$, $p = 0.068$)—unsurprising, since they share a base model—while `phi3` is essentially disjoint ($D = 0.901$).

R. Summary of statistical procedures

Table IX consolidates every inferential procedure reported in this paper: what each measures, why that test was chosen over alternatives, and what it returned. We include the tests that produced null and negative results, and the one test that could not be computed at all, since selectively reporting only significant procedures is itself a source of bias.

V. DISCUSSION

A. What the human anchor changes

The practical question facing an engineering team is not whether generated IaC contains findings—all IaC does—but whether adopting a model changes their risk relative to writing it themselves. Model-only benchmarks cannot answer this. Our size-matched comparison gives a concrete figure: roughly a $3.5\times$ increase in findings per declared resource, consistent across every vendor tested.

The consistency is the surprising part. Configurations spanning four vendors, open and closed weights, three reasoning modes, and a $10\times$ range in schema-validity pass rate nonetheless fall within a $3.21\text{--}3.87\times$ band. This argues against a model-selection remedy: choosing a different frontier model does not meaningfully change the security profile of the generated infrastructure. It also argues that the gap originates upstream of any particular vendor’s alignment work—plausibly in the training distribution, which is dominated by public repository code where permissive defaults are common [1].

B. The gap is largest where review is weakest

The ratio declines monotonically with artifact size, from $4.9\times$ on single-resource templates to a non-significant $1.4\times$ on templates declaring twenty or more resources. This inverts the intuition that complex generations are the risky ones.

The practical implication is uncomfortable. Small generated snippets are exactly the artifacts least likely to receive careful review—a five-line storage bucket definition reads as obviously correct—yet they carry the largest relative security penalty. Large generated architectures attract scrutiny and, per resource, are closer to human-authored code. Review effort is therefore misallocated by default: it scales with artifact size, while relative risk scales inversely.

C. Reasoning modes are not a solution, but they are not nothing

Extended thinking produces a real, statistically significant reduction (-13.2%), significantly outperforms the prompt-based substitute (-12.0% , $p = 0.0013$), and moves no configuration near the human baseline. A 13% reduction against a $3.5\times$ deficit does not close the gap.

The token instrumentation suggests why, and also where the ceiling might not be. Because the model spends under 1% of its output budget on reasoning for this task class, the observed effect is produced by a barely-engaged mechanism. Two readings are consistent with our data: either IaC generation does not elicit reasoning because it is perceived as a recall task rather than a problem-solving one, or current reasoning training does not transfer to declarative configuration. Distinguishing these requires interventions that force higher reasoning expenditure, which is a natural follow-up.

For practitioners the immediate guidance is narrower and firmer: if a reasoning mode is available, enabling it is measurably better than instructing the model to “think step by step,” and the latter provides no measurable security benefit at all.

D. Absolute counts mislead

Three results in this study point the same way. `phi3` records the fewest absolute findings and the highest per-resource rate. Severity-tiered counts rank the most capable configuration worst, purely because it emits the most code. Cross-format densities are dominated by a parser-dependent denominator. Each is a case where the intuitive metric inverts the correct one.

We therefore recommend that IaC security benchmarks report exposure-normalized rates with an explicit denominator definition, and treat absolute counts as descriptive only. The negative binomial model with a $\log(\text{resources})$ offset is the form we found necessary: overdispersion of $130\times$ rules out Poisson, and omitting the offset converts a rate comparison into a code-volume comparison.

E. Implications for evaluation methodology

Two methodological results generalize beyond IaC. First, complete-case Friedman is uncomputable on a realistic multi-model design: with twelve configurations, no scenario had complete coverage. Benchmarks that lose cells to refusals, quota exhaustion, or truncation—which is most of them at scale—should default to incomplete-block statistics such as Skillings–Mack rather than silently discarding blocks.

Second, our LLM judge is substantially reliable for factual verification (hallucination $\kappa = 0.640$) and near-useless for architectural judgment ($\kappa = 0.177$). Given the rapid adoption of model-based evaluation [23], [24], per-criterion calibration against human raters should be reported rather than assumed. The near-chance inter-human agreement on security-test relevance ($\kappa = 0.059$) further suggests that some evaluation criteria may not be reliably measurable by humans either, and that disagreement should be reported rather than adjudicated away.

TABLE IX

ALL STATISTICAL PROCEDURES PERFORMED, WHAT EACH MEASURES, AND ITS OUTCOME. “→” GIVES THE INTERPRETATION. TESTS ARE GROUPED BY THE QUESTION THEY ANSWER; NULL AND NEGATIVE RESULTS ARE INCLUDED.

Test	What it measures	Why this test	Result and outcome
Q1. Are LLMs less secure than human engineers?			
Spearman rank correlation	Association between vulnerability density and artifact size	Non-parametric; densities are skewed and the relationship need not be linear	$\rho = -0.546$, $p = 1.7 \times 10^{-78}$ → density falls sharply with size, so <i>unmatched</i> comparisons measure artifact size, not security. Motivates all matching below.
Mann-Whitney U (per size stratum)	LLM vs. human density within matched resource-count bins	Two independent samples, non-normal; matching removes the size confound	Significant in 5 of 6 bins ($p < 0.05$); ratio $4.9 \times$ (1 resource) → $1.4 \times$ (20+, n.s.) → gap is real and largest on the simplest artifacts .
Aggregate matched ratio	Per-configuration density relative to human, weighted across bins	Summarizes the stratified comparison in one figure per configuration	All 12 configurations in $3.21 \times -3.87 \times$ → consistent across vendors, weights, and reasoning modes .
Q2. Do the configurations differ from each other?			
Skellings-Mack	Omnibus equality of configurations across scenarios (blocked design)	Generalizes Friedman to <i>incomplete</i> blocks; required here because no scenario has all 12 configurations present	Simple: $\chi^2 = 69.3$, $df = 10$, $p = 6.0 \times 10^{-11}$ (60 blocks). Complex: $\chi^2 = 81.2$, $df = 11$, $p = 8.7 \times 10^{-13}$ (40 blocks) → configurations differ .
Friedman (complete-case)	Same hypothesis, listwise deletion	Classical alternative; reported to justify its replacement	Uncomputable : 0 usable blocks in both strata → negative methodological result (Section IV-P).
Wilcoxon signed-rank + Holm	Pairwise configuration differences after a significant omnibus	Paired (same scenarios); Holm controls family-wise error over 66 pairs	10/66 significant (simple), 9/66 (complex) → cross-vendor gaps dominate; no same-model reasoning-arm pair survives correction .
Q3. Does reasoning improve security?			
Wilcoxon signed-rank (paired contrasts)	Within-model effect of each reasoning mode on density	Same model, same scenarios, one variable toggled; non-normal paired data	Extended thinking -13.2% ($p = 0.012$); CoT→thinking -12.0% ($p = 0.0013$); prompted CoT alone -1.3% (n.s.) → vendor reasoning helps modestly; prompted CoT does not .
Reasoning-token share	Fraction of output budget spent on reasoning	Direct instrumentation rather than inference	$<1\%$ of output tokens → explains why the effect is bounded.
Q4. What drives the rate, controlling for exposure?			
Negative binomial GEE (log exposure offset)	Per-resource finding rate by configuration and complexity	Counts are overdispersed; offset makes coefficients <i>rates</i> not volumes; GEE handles repeated scenarios	Converged with exchangeable correlation → IRRs interpretable per resource.
Overdispersion check	Variance-to-mean ratio of the count outcome	Determines whether Poisson is admissible	$\approx 130 \times$ → Poisson rejected; NB2 required (Section VI).
Configuration $\times$ complexity interaction	Whether a configuration’s rate shifts disproportionately between strata	Tests differential degradation directly	5 of 11 significant, all $IRR < 1$ → lower simple-stratum rates once exposure is controlled; the previously reported “complexity relaxation” effect does not survive ($p = 0.275$).
Q5. Is generated IaC structurally like human IaC?			
Two-sample Kolmogorov-Smirnov (vs. human)	Distributional equality on AST depth, resource count, diversity	Distribution-free; compares whole distributions, not just means	36/36 tests reject ($p < 0.05$) → all configurations structurally diverge ; gpt-4o closest ($D = 0.160$), phi3 most divergent ($D = 0.990$).
KS (configuration vs. field)	Whether each configuration occupies a distinct structural niche	Same rationale, applied pairwise against the pooled remainder	30/48 significant → configurations are structurally distinguishable.
KS (simple vs. complex)	Whether the complexity strata actually differ in the artifacts	Validates the stratification rather than assuming it	Rejects equality → stratification is real , not just prompt wording.
Q6. Is the evaluation itself trustworthy?			
Fleiss’ κ	Agreement among three human raters per criterion	Chance-corrected agreement for >2 raters	Plausibility 0.391; hallucination 0.266; coherence 0.210; security relevance 0.059 (near chance) → some criteria are not reliably measurable by humans.
Cohen’s κ (binary)	Human consensus vs. LLM judge on the hallucination flag	Chance-corrected agreement for a binary label	$\kappa = 0.640$, 94.4% exact → judge is substantially reliable for factual verification .
Quadratic-weighted κ (ordinal)	Human consensus vs. LLM judge on the 1–5 criteria	Penalizes large ordinal disagreements more than adjacent ones	Plausibility 0.795; relevance 0.489; coherence 0.177 → judge is unreliable for architectural judgment.
Q7. Negative result: the validity–security paradox			
Pearson correlation	Schema-validity pass rate vs. vulnerability density across configurations	Tests the prior hypothesis that more deployable models are more vulnerable	$r = 0.158$, $p = 0.625$ → hypothesis unsupported ; reported as a negative result.

VI. THREATS TO VALIDITY

Construct. The human corpus is *not* a matched control: those templates were not written against our scenarios, and many are curated examples rather than production infrastructure. Example templates may be deliberately minimal, which could bias the baseline in either direction; we do not claim to know which. Vendor reasoning mechanisms also differ (token budget versus effort level) and are not pooled.

Internal. Scanner findings indicate policy deviations, not proven exploitability; “zero findings” means only that three rulesets flagged nothing. Checkov does not emit severity tiers without a commercial subscription, so severity analysis reflects Trivy and KICS only. Scenarios were authored by a model that also appears under test, an unquantified residual risk mitigated by requirements-only phrasing and independent human and judge review.

Measurement infrastructure. During this study we identified several defects that produced plausible-looking but incorrect results with no error raised: a platform encoding mismatch that aborted scanning for an entire configuration; a filename-handling rule that silently discarded one engine’s reports for configurations whose names contain a period; a schema mismatch that caused one engine’s findings to parse as zero for ten of twelve configurations; a resource-count source that degraded to a constant denominator, inflating an apparent effect to +2528% where the corrected value is −14%; and ungenerated scenarios being scored as zero-vulnerability. Each would have survived review because the resulting numbers looked reasonable. We report this explicitly: in multi-tool benchmarks, derived counts must be reconciled against source artifacts rather than trusted, and we release coverage manifests and regeneration scripts so that any published figure can be recomputed.

External. Hosted models change without notice; results describe the configurations as accessed. Terraform dominates the corpus, and engine rule coverage differs by format. Statistical power for the paired reasoning contrasts is limited ($n \leq 60$), and only two model families expose reasoning arms.

VII. CONCLUSION AND FUTURE WORK

Measured against human-authored infrastructure through an identical toolchain and matched on artifact size, current language models produce IaC with approximately $3.2\times$ – $3.9\times$ the vulnerability density of human engineers, with remarkable consistency across vendors and weight availability. The gap is widest on the simplest tasks. Vendor reasoning modes deliver a real but modest improvement and meaningfully outperform prompted chain-of-thought, which alone provides no measurable security benefit—though the mechanism is barely engaged on this task class, consuming under 1% of output tokens.

Future work should pursue a matched human control authored against the same scenarios; larger samples and additional model families for the reasoning contrasts; a sensitivity sweep over reasoning-budget settings; exploitability triage to

connect policy findings to realizable attack paths; deployment-time validation beyond static analysis; and improved rubric anchoring, given that expert agreement on security-test relevance was near chance.

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We thank the three practicing cloud and security engineers who independently reviewed the stratified scenario sample that underpins Section IV-O. They scored every sampled scenario without seeing one another’s ratings, and their disagreement—not merely their agreement—shaped the reliability analysis and the limits we place on the LLM judge. All three consented to be named:

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Their raw scores are released in anonymized form (R1–R3); the mapping between identifiers and identities is retained only by the author and is not distributed.

DATA AVAILABILITY

The pipeline, statistical analysis code, figure generation, coverage manifests, and threats-to-validity documentation are released. Every figure and table is regenerated from the released data by a single command. The generated corpus, raw scanner output, human reference corpus, and all derived result tables are archived at huggingface.co/datasets/AnimeshShaw/GenIaC-SecBench; the code is at github.com/AnimeshShaw/GenIaC-SecBench.

REFERENCES

- [1] A. Rahman, C. Parnin, and L. Williams, “The seven sins: Security smells in infrastructure as code scripts,” in *IEEE/ACM International Conference on Software Engineering (ICSE)*, 2019, pp. 164–175.
- [2] A. Verdet, M. Hamdaqa, L. Da Silva, and F. Khomh, “Exploring security practices in infrastructure as code: An empirical study,” *Empirical Software Engineering*, 2023.
- [3] H. Pearce, B. Ahmad, B. Tan, B. Dolan-Gavitt, and R. Karri, “Asleep at the keyboard? assessing the security of GitHub Copilot’s code contributions,” in *IEEE Symposium on Security and Privacy (S&P)*, 2022, pp. 754–768.
- [4] G. Sandoval, H. Pearce, T. Nys, R. Karri, S. Garg, and B. Dolan-Gavitt, “Lost at C: A user study on the security implications of large language model code assistants,” in *USENIX Security Symposium*, 2023, pp. 2205–2222.
- [5] N. Tihanyi, T. Bisztray, M. A. Ferrag, R. Jain, and L. C. Cordeiro, “How secure is AI-generated code: A large-scale comparison of large language models,” *Empirical Software Engineering*, 2025.
- [6] N. Perry, M. Srivastava, D. Kumar, and D. Boneh, “Do users write more insecure code with AI assistants?” in *ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2023, pp. 2785–2799.
- [7] G. Vargas, R. B. Mansilha, and D. Kreutz, “Security-first evaluation of text-to-Terraform generation,” *arXiv preprint arXiv:2608.02672*, 2026, accepted at SBSeg 2026.
- [8] J. H. Skillings and G. A. Mack, “On the use of a Friedman-type statistic in balanced and unbalanced block designs,” *Technometrics*, vol. 23, no. 2, pp. 171–177, 1981.

- [9] M. L. Siddiq and J. C. S. Santos, "SecurityEval dataset: Mining vulnerability examples to evaluate machine learning-based code generation techniques," in *International Workshop on Mining Software Repositories Applications for Privacy and Security (MSR4P&S)*, 2022, pp. 29–33.
- [10] J. Liu, C. S. Xia, Y. Wang, and L. Zhang, "Is your code generated by ChatGPT really correct? rigorous evaluation of large language models for code generation," *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [11] N. Saavedra and J. F. Ferreira, "GLITCH: Automated polyglot security smell detection in infrastructure as code," in *IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2023.
- [12] A. Rahman, R. Mahdavi-Hezaveh, and L. Williams, "A systematic mapping study of infrastructure as code research," *Information and Software Technology*, vol. 108, pp. 65–77, 2019.
- [13] Prisma Cloud / Bridgecrew, "Checkov: Policy-as-code for infrastructure as code," <https://github.com/bridgecrewio/checkov>, 2026.
- [14] Aqua Security, "Trivy: Comprehensive security scanner," <https://github.com/aquasecurity/trivy>, 2026.
- [15] Checkmarx, "KICS: Keeping infrastructure as code secure," <https://github.com/Checkmarx/kics>, 2026.
- [16] Center for Internet Security, "CIS benchmarks," <https://www.cisecurity.org/cis-benchmarks>, 2026.
- [17] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. Le, and D. Zhou, "Chain-of-thought prompting elicits reasoning in large language models," *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [18] T. Kojima, S. S. Gu, M. Reid, Y. Matsuo, and Y. Iwasawa, "Large language models are zero-shot reasoners," *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [19] Z. Sprague, F. Yin, J. D. Rodriguez *et al.*, "To CoT or not to CoT? chain-of-thought helps mainly on math and symbolic reasoning," *International Conference on Learning Representations (ICLR)*, 2025.
- [20] OpenAI, "Learning to reason with LLMs," *OpenAI Technical Report*, 2024.
- [21] DeepSeek-AI, "DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning," *arXiv preprint arXiv:2501.12948*, 2025.
- [22] Anthropic, "Extended thinking," <https://docs.anthropic.com/en/docs/build-with-claude/extended-thinking>, 2026.
- [23] L. Zheng, W.-L. Chiang, Y. Sheng *et al.*, "Judging LLM-as-a-judge with MT-Bench and Chatbot Arena," *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [24] J. Gu, X. Jiang, Z. Shi *et al.*, "A survey on LLM-as-a-judge," *arXiv preprint arXiv:2411.15594*, 2025.
- [25] A. Panickssery, S. R. Bowman, and S. Feng, "LLM evaluators recognize and favor their own generations," *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [26] P. Ralph *et al.*, "Empirical standards for software engineering research," *arXiv preprint arXiv:2010.03525*, 2021.
- [27] M. Friedman, "The use of ranks to avoid the assumption of normality implicit in the analysis of variance," *Journal of the American Statistical Association*, vol. 32, no. 200, pp. 675–701, 1937.
- [28] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945.
- [29] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.
- [30] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *Journal of Machine Learning Research*, vol. 7, pp. 1–30, 2006.
- [31] K.-Y. Liang and S. L. Zeger, "Longitudinal data analysis using generalized linear models," *Biometrika*, vol. 73, no. 1, pp. 13–22, 1986.
- [32] J. M. Hilbe, *Negative Binomial Regression*, 2nd ed. Cambridge University Press, 2011.
- [33] H. B. Mann and D. R. Whitney, "On a test of whether one of two random variables is stochastically larger than the other," *Annals of Mathematical Statistics*, vol. 18, no. 1, pp. 50–60, 1947.
- [34] F. J. Massey Jr., "The Kolmogorov-Smirnov test for goodness of fit," *Journal of the American Statistical Association*, vol. 46, no. 253, pp. 68–78, 1951.
- [35] J. L. Fleiss, "Measuring nominal scale agreement among many raters," *Psychological Bulletin*, vol. 76, no. 5, pp. 378–382, 1971.
- [36] J. R. Landis and G. G. Koch, "The measurement of observer agreement for categorical data," *Biometrics*, vol. 33, no. 1, pp. 159–174, 1977.
- [37] J. Cohen, "A coefficient of agreement for nominal scales," *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, 1960.